Data Technician Workbook
Week 3
Databases and SQL

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Day 1: Task 1

Scenario-Based SQL Activity: Designing a Normalised Database

Business Scenario

You have been hired as a **junior data technician** for a small training company called **SkillUp Academy**.
The company currently stores all its data in a **single spreadsheet**, which is causing:

- Duplicate information
- Data inconsistencies
- Difficulty updating records

Your task is to **design a simple relational database** that organises this data more effectively.

Task 1: Database Purpose

Describe the purpose of the database in your own words. Include:

- What the database will be used for
- Who will use it
- What type of data it will store

✍️ *Write in full sentences.*

The purpose of this database will be to store and organise information in a structured way so that it can be managed, updated, and seen efficiently and easily. The database will help the users keep track of important records and reduce errors. The database will be used for managing common information seen within businesses such as customers, products, orders, employees and financial records dependent on the businesses needs/requirements. people who will use the database will be employees, managers, and administrators. customers may also access limited and nonsensitive records. Investors and shareholders may also access financial records of the company which will be required if the company is publicly traded. Overall the database aims to improve organisation, accuracy, efficiency and security of sensitive data

Part 2: Identify Tables and Columns

Task 2: Identify Tables

From the scenario, identify **at least four tables** that should be created.

For each table:

- Give the **table name**
- List the **columns**
- Identify the **primary key**

✍️ *You may present your answer as a list or table.*

Table name – customers , orders, products, employees

Columns

1. Customers... stores customer information – customer id , first and last name, email, phone number
2. Orders... stores order information , and amount – order id , customer id, order date, total amount ordered
3. Products... will store stocktake and name of the product along with prices – product id, product name, price, stock quantity
4. Employees... stores sensetive internal information suhc as roles salaries and names. – salary , first and last name, job role, an unique employee id

Part 3: Define Relationships Between Tables

Task 3: Table Relationships

Explain how your tables are connected.

You must:

- Identify **one-to-many relationships**

- Explain which columns are used as **foreign keys**
- Explain why these relationships are needed

The tables are connected using one to many relationships, where one record in the table can be linked to multiple records in another table. An example of this is one customer can place many orders and one employee can manage many orders. The database will use foreign keys to create these connections between tables. For example within an order table the customer id will be a foreign key which likes to the customers table. This relationship is important as it will help keep the database organised and reduce duplicate data. It wil also improve the accuracy and make it easier to update or search the database.

Part 4: Reflection (Optional / Stretch)

Task 5: Reflection

Answer the following:

- What problems does this database design solve?
- What issues might occur if all data stayed in one table?

Problems that the database design solves. Duplicated data, organisational issues , difficulty with managing large amounts of information, searching the databse and storing information securely. Issues that could occur if all data stayed in one table is that there could be repeated information and make it harder to manage in general as the data grows larger.

Day 1: Task 2

Choosing the Right Type of Database
Business Scenario
SkillUp Academy is planning to expand its systems.
In addition to structured course and learner data, the company now wants to store:

- Website click data
- User activity logs
- Feedback comments from learners
- Uploaded documents

They are unsure whether a **relational database** or a **non-relational database** would be more suitable for this data.
You have been asked to help explain the options.

Task 1: Compare Database Types
In your own words, explain the difference between:

- A **relational database**
- A **non-relational (NoSQL) database**

Your answer should include:

- How data is stored
- How structured the data is
- An example use case for each

✍️ *Write in clear, simple language.*

A relational database will store data in tables that have rows and columns where the tables are connected using relationships. The data is very organised and structured compared to non-relational databases, which in turn will make it great for banks or online shops. Online shops for example can have a table for customers and another for orders which are linked together by a foreign key such as a customer ID.

Non- relational databases store data more flexibly such as with documents or collections rather than tables. This is useful alternate way to store big data as data changes often or does not fit within the parameters of rows and columns. An example of this type of database being used is social media apps which will store all sorts of data such as posts, comments, and pictures.

Task 2: Identify Suitable Data for a Non-Relational Database

From the scenario above, identify **two types of data** that would be better stored in a **non-relational database**.

Examples may include:

- User activity logs
- Feedback text
- Website events

For each type of data:

- State **what the data is**
- Explain **why a non-relational database is suitable**

✍️ *Focus on flexibility, structure, and data format.*

1. The first type of data which would be better stored in a non-relational database could be user login times for software, apps, or games. The data which it would include would be the times , actions completed on a webiste and pages visitied. This may not apply to games. It would be suitable as the data strucute would change frequently based on login times

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Task 3: Justify Your Choice

Explain **why this data would not be ideal** for a traditional relational database.

Consider:

- Data structure
- Frequency of changes
- Volume of data

✍️ *Use plain English — no advanced terminology required.*

The user login activity logs and customer feedback from online shops wouldn't be ideal for a traditional relational database due to the fact that the data can change very quickly and often. It will not always follow the same structure. An example is that customer feedback could be either short or long and include emojis as well as text which makes it difficult to organise into columns and rows due to each response being abstract. A relational database could become slow and hard to navigate to so much data.

Day 2: Task 1

You are a junior data analyst at **Northwind Traders**. Your manager has assigned you a series of real-world business tasks. For each scenario:

1. **Write an SQL query** to solve the request. Write an SQL query to solve the request.
2. Run the query in MySQL Workbench using the Northwind database.
3. Paste your query in the box below each question as evidence of completion.

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1. 📘 *Retrieve Full Customer Data*

Your manager has asked you to export the full list of all customer details into a report.

➡️ **Write a SQL query to retrieve all columns from the Customers table.**

```
select *

FROM Customers;
```

---

2. 📘 *Customer Names and Cities for Marketing*

A marketing analyst is targeting a campaign based on customer names and their cities.

➡️ **Write a SQL query to retrieve only the CustomerName and City columns from the Customers table.**

```
select CustomerName, City

FROM Customers;
```

---

3. 📘 *Unique Cities for Delivery Network Expansion*

The logistics team wants to know all the different cities where customers are located (no duplicates).

➡️ **Write a SQL query to retrieve distinct values from the City column in the Customers table.**

```
select DISTINCT City
FROM Customers;
```

---

4. 📘 *High-Value Products Report*

The product manager wants to analyse products priced over £50.

➡️ **Write a SQL query to retrieve all columns from the Products table where the Price is greater than 50.**

```
select *
FROM Products
WHERE Price > 50;
```

---

5. 📘 *International Customers Targeting (USA & UK)*

The marketing team is preparing campaigns for customers in the USA and UK.

➡️ **Write a SQL query to retrieve all columns from the Customers table where the Country is either 'USA' or 'UK'.**

```
select *
FROM Customers
WHERE Country IN ('USA', 'UK');
```

---

6. 📘 *Recent Orders Report*

Your team needs to analyze the latest orders.

➡️ **Write a SQL query to retrieve all columns from the Orders table, sorted by OrderDate in descending order.**

```
select *
FROM Orders
ORDER BY OrderDate DESC;
```

---

7. 📘 *Mid-Range Products Listing*

You are preparing a product list for items priced between £20 and £50.

➡️ **Write a SQL query to retrieve all columns from the Products table where the Price is between 20 and 50, ordered by descending Price.**

```
SELECT *

FROM Products

WHERE Price BETWEEN 20 AND 50

ORDER BY Price DESC;
```

---

8. 📘 *Local Marketing in the US (Portland & Kirkland)*

You're focusing on customers from Portland and Kirkland in the USA.

➡️ **Write a SQL query to retrieve all columns from the Customers table where the Country is 'USA' and City is either 'Portland' or 'Kirkland', ordered by ascending CustomerName.**

```
SELECT *

FROM Customers

WHERE Country = 'USA'

AND City IN ('Portland', 'Kirkland')

ORDER BY CustomerName ASC;
```

9. 📘 *Customers from the UK or London*
Prepare a list for promotional emails targeting customers from the UK or those based in London.
➡️ **Write a SQL query to retrieve all columns from the Customers table where the Country is 'UK' or City is 'London', ordered by descending CustomerName.**

```
select*
FROM Customers

WHERE Country = 'UK'

OR City = 'London'

ORDER BY CustomerName DESC;
```

10. 📘 *Product Inventory for Selected Categories*
The inventory team needs products from Category 1 or 2, sorted alphabetically.
➡️ **Write a SQL query to retrieve all columns from the Products table where the CategoryID is 1 or 2, ordered by ascending ProductName.**

```
select *
FROM Products

WHERE CategoryID IN (1,2)

ORDER BY ProductName ASC;
```

Day 3: Task 1

Scenario-Based Activity: Understanding SQL JOINs

Business Scenario

SkillUp Academy stores its data across several related tables, such as:

- Learners
- Courses
- Trainers
- Enrolments

To create reports (for example, seeing which learners are enrolled on which courses), the data team needs to **combine data from multiple tables**.

Different types of **SQL JOINs** are used depending on what information is needed.

Task 1: Explain JOIN Types (Plain English)

For each JOIN type below:

- Explain **what the JOIN does** in simple terms
- Explain **when it would be used** in a real scenario

You do **not** need to write SQL code unless instructed.

JOIN Types to Explain

- Inner Join
- Left Join
- Right Join
- Full Join
- Cross Join

✍️ *Use short, clear explanations.*

Inner join – returns only records that match on both tables , an example would be learners who are enrolled ona course and learners without enrolments will not appear.

Left join. all records from the left table and matches records from the right table. If there is not a match then Null will appear. An example would be to show all learners including those not yet enrolled in a class

Right join. Returns all records from right table and matching records from left table. Again if there is not a match null will appear. For a real world example it could show all course even if there are no learners on that course

Full join. Returns all matching records from both tables, and also unmatched records from both sides. This can be shown as an example that every learner and every course is shown as well as learners who are not enrolled into a course or courses with no students

Cross join, this will combine every row from none table with every row from another table. An example in college administration is that it could create every possible learner course combination if the admin team wanted to prepare and plan

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Task 2: Match JOINs to Data Scenarios

Below are example reporting needs at SkillUp Academy.

For each one:

- Identify **which JOIN type** would be most suitable
- Explain **why**

Example Scenarios

1. Viewing only learners who are enrolled on a course
2. Listing all courses, including those with no enrolments yet
3. Showing all learners and all courses, even if they are not linked
4. Creating all possible combinations of learners and courses for planning
5. Comparing learners to other learners (e.g. same course, same city)
6. Viewing enrolments even if some related data is missing

✍️ *Explain your reasoning in plain English.*

Q1, INNER JOIN

This will only show that learners that have matching enrolment records. If a learner is not enrolled, they will not appear.

2. Left join, as this will keep al course results regardless of no enrolments on the course for learners

3. Full outer join, this will show every learner and every course. It will not matter whether there is a matching enrolment between them
4. Cross join, this will combine every learner with every course to create all mathematically possible combinations on the table
5. Self join, as it will join a table to itself so records in the same table can be compared when analysing the data
6. Right join, will allow enrolment records to be able to still appear even inf there are some pupil or course information missing in the records

Day 3: Task 2

You are a junior data analyst at **Northwind Traders**. Your manager has assigned you a series of real-world business tasks. For each scenario:

1. **Write an SQL query** to solve the request. Write an SQL query to solve the request.
2. Run the query in MySQL Workbench using the Northwind database.
3. Paste your query in the box below each question as evidence of completion.

1. 📘 *Linking Products to Suppliers*

Management wants to know which supplier provides each product in the inventory.

➡️ **Write a SQL query to find the supplier of each product.**

```
SELECT Products.ProductName, Suppliers.SupplierName

FROM Products

INNER JOIN Suppliers

ON Products.SupplierID = Suppliers.SupplierID;
```

---

2. 📘 *Classifying Products by Category*

The category manager is reviewing how products are organized.

➡️ **Write a SQL query to find the category of each product.**

```
SELECT Products.ProductName, Categories.CategoryName

FROM Products

INNER JOIN Categories

ON Products.CategoryID = Categories.CategoryID;
```

---

3. 📘 *Category-Specific Product Report: Meat/Poultry*

The food department wants to view all items in the *Meat/Poultry* category.

➡️ **Write a SQL query to retrieve all products belonging to the Meat/Poultry category.**

```
SELECT Products.ProductName, Categories.CategoryName

FROM Products

INNER JOIN Categories

ON Products.CategoryID = Categories.CategoryID

WHERE Categories.CategoryName = 'Meat/Poultry';
```

---

4. 📘 *Complete Order Overview*

The business team wants to see a detailed order list with customer and employee information.

➡️ **Write a SQL query to retrieve the Order ID, Order Date, Customer Name, and Employee Name for all orders.**

```
SELECT Orders.OrderID, Orders.OrderDate, Customers.CustomerName, Employees.FirstName, Employees.LastName

FROM Orders

INNER JOIN Customers

ON Orders.CustomerID = Customers.CustomerID

INNER JOIN Employees

ON Orders.EmployeeID = Employees.EmployeeID;
```

---

5. 📘 *Supply Chain Overview Report*
Your manager wants to see the product name, its category, and the name of its supplier all in one report.
➡️ **Write a SQL query to retrieve the Product Name, Category Name, and Supplier Name for all products.**

```
SELECT Products.ProductName, Categories.CategoryName, Suppliers.SupplierName

FROM Products

INNER JOIN Categories

ON Products.CategoryID = Categories.CategoryID

INNER JOIN Suppliers

ON Products.SupplierID = Suppliers.SupplierID;
```

---

6. 📘 *Yearly Order Summary – 1996*
The team is auditing customer orders made in 1996.
➡️ **Write a SQL query to create a report for all the orders of 1996 and their customers.**

```
SELECT Orders.OrderID, Orders.OrderDate, Customers.CustomerName

FROM Orders

INNER JOIN Customers

ON Orders.CustomerID = Customers.CustomerID

WHERE YEAR(Orders.OrderDate) = 1996;
```

7. 📘 *Product Count by Category*

The product team wants to know how many products exist under each category.

➡️ **Write a SQL query to retrieve all categories along with the number of products in each category.**

```
SELECT Categories.CategoryName, COUNT(Products.ProductID) AS ProductCount

FROM Categories

LEFT JOIN Products

ON Categories.CategoryID = Products.CategoryID

GROUP BY Categories.CategoryName;
```

---

8. 📘 *Sales Volume Breakdown*

The sales department wants to analyze how much of each product was ordered and at what price.

➡️ **Write a SQL query to retrieve all products with their prices and the quantity ordered for each product.**

```
SELECT Products.ProductName, Products.Price, OrderDetails.Quantity

FROM Products

INNER JOIN OrderDetails

ON Products.ProductID = OrderDetails.ProductID;
```

Day 4: Task 1: SQL Practical

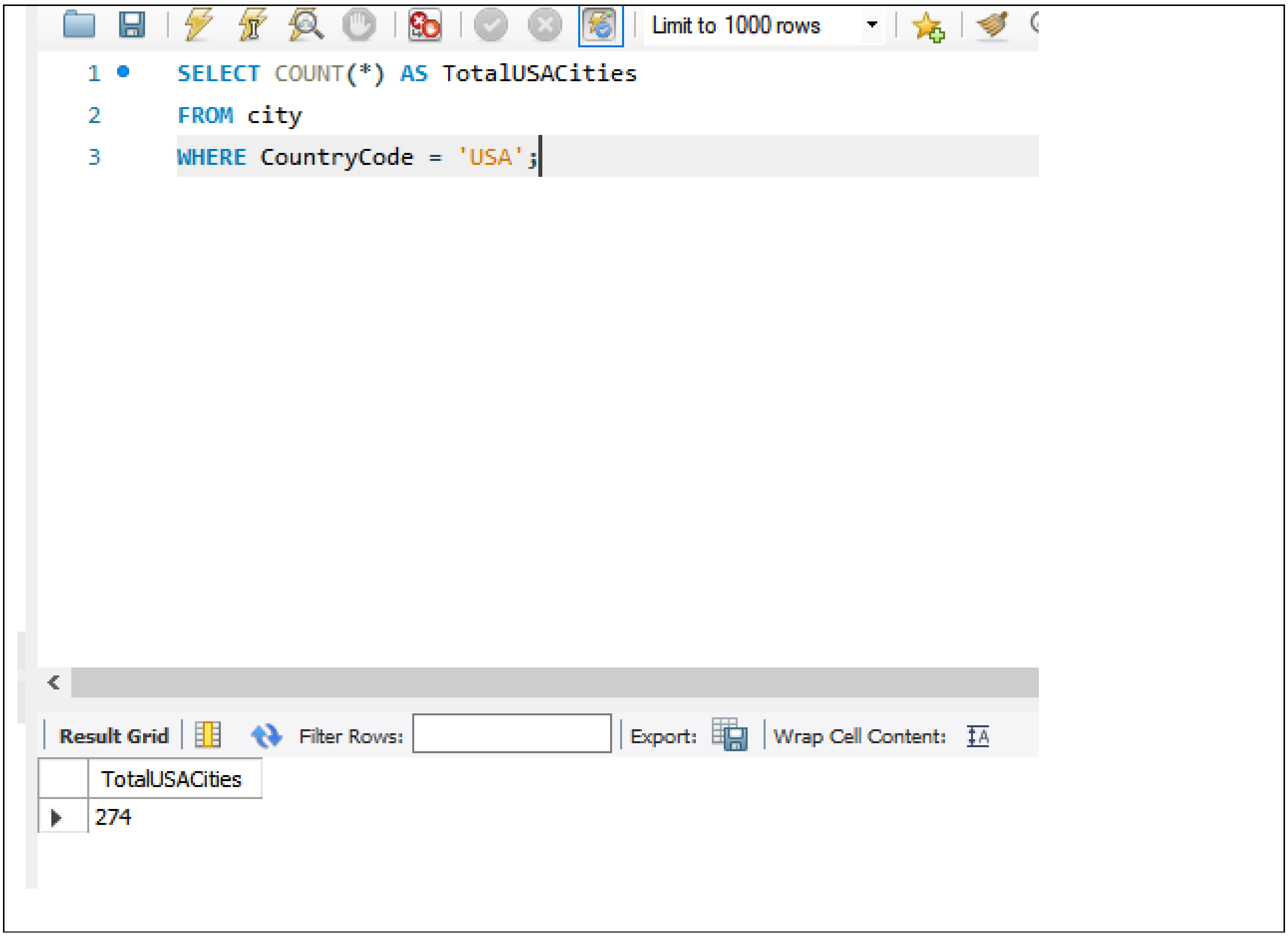
You are working as a **junior data analyst** for an international research organization analysing countries, cities, and languages from around the world. Your job is to write and run SQL queries that provide useful global insights.
For each task:

1. **Write the appropriate SQL query** to meet the scenario's objective.
2. **Run your query** using **MySQL Workbench** and the **world** database.
3. **Take a screenshot** that clearly shows both your SQL code and the result table.
4. **Paste the screenshot in the box provided for each question as evidence of completion.**

Setting up the database:

1. **Download world_db here**
2. **Follow each step to create your database here**
3. **Create Schema Design Diagram. How to create Schema Design.webm**

1. **Count Cities in USA:** *Scenario:* You've been tasked with conducting a demographic analysis of cities in the United States. Your first step is to determine the total number of cities within the country to provide a baseline for further analysis.



2. **Country with Highest Life Expectancy:** *Scenario:* As part of a global health initiative, you've been assigned to identify the country with the highest life expectancy. This information will be crucial for prioritising healthcare resources and interventions.

```sql
SELECT Name, LifeExpectancy
FROM country
ORDER BY LifeExpectancy DESC
LIMIT 1;
```

sult Grid Filter Rows: Expo

Name	LifeExpectancy
Andorra	83.5

3.

3. **"New Year Promotion: Featuring Cities with 'New :** *Scenario:* In anticipation of the upcoming New Year, your travel agency is gearing up for a special promotion featuring cities with names including the word 'New'. You're tasked with swiftly compiling a list of all cities from around the world. This curated selection will be essential in creating promotional materials and enticing travellers with exciting destinations to kick off the New Year in style.

```sql
SELECT Name, CountryCode, Population
FROM city
WHERE Name LIKE '%New%';
```

Result Grid | Filter Rows: | Export: | Wra

Name	CountryCode	Population
Newcastle	AUS	270324
Newcastle upon Tyne	GBR	189150
Newport	GBR	139000
Newcastle	ZAF	222993
Kowloon and New Kowloon	HKG	1987996
New Bombay	IND	307297
New Delhi	IND	301297
Khanewal	PAK	133000
New York	USA	8008278
New Orleans	USA	484674
Newark	USA	273546
Newport News	USA	180150
New Haven	USA	123626
New Bedford	USA	94780

4. **Display Columns with Limit (First 10 Rows):** *Scenario:* You're tasked with providing a brief overview of the most populous cities in the world. To keep the report concise, you're instructed to list only the first 10 cities by population from the database.

```sql
SELECT Name, CountryCode, Population
FROM city
ORDER BY Population DESC
LIMIT 10;
```

Name	CountryCode	Population
Mumbai (Bombay)	IND	10500000
Seoul	KOR	9981619
São Paulo	BRA	9968485
Shanghai	CHN	9696300
Jakarta	IDN	9604900
Karachi	PAK	9269265
Istanbul	TUR	8787958
Ciudad de México	MEX	8591309
Moscow	RUS	8389200
New York	USA	8008278

5. **Cities with Population Larger than 2,000,000:** *Scenario:* A real estate developer is interested in cities with substantial population sizes for potential investment opportunities. You're tasked with identifying cities from the database with populations exceeding 2 million to focus their research efforts.

```sql
SELECT Name, CountryCode, Population
FROM city
WHERE Population > 2000000
ORDER BY Population DESC;
```

Name	CountryCode	Population
Mumbai (Bombay)	IND	10500000
Seoul	KOR	9981619
São Paulo	BRA	9968485
Shanghai	CHN	9696300
Jakarta	IDN	9604900
Karachi	PAK	9269265
Istanbul	TUR	8787958

6. **Cities Beginning with 'Be' Prefix:** *Scenario:* A travel blogger is planning a series of articles featuring cities with unique names. You're tasked with compiling a list of

cities from the database that start with the prefix 'Be' to assist in the blogger's content creation process.

```sql
SELECT Name, CountryCode, Population
FROM city
WHERE Name LIKE 'Be%';
```

Name	CountryCode	Population
Béjaïa	DZA	117162
Béchar	DZA	107311
Benguela	AGO	128300
Berazategui	ARG	276916
Belize City	BLZ	55810
Belmopan	BLZ	7105
Belo Horizonte	BRA	2139125
Belém	BRA	1186926
Belford Roxo	BRA	425194
Betim	BRA	302108
Bento Gonçal...	BRA	89254
Belfast	GBR	287500

7. **Cities with Population Between 500,000-1,000,000:** *Scenario:* An urban planning committee needs to identify mid-sized cities suitable for infrastructure development projects. You're tasked with identifying cities with populations ranging between 500,000 and 1 million to inform their decision-making process.

```sql
SELECT Name, CountryCode, Population
FROM city
WHERE Population BETWEEN 500000 AND 1000000
ORDER BY Population DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Name	CountryCode	Population
Amman	JOR	1000000
Mogadishu	SOM	997000
Volgograd	RUS	993400
Sendai	JPN	989975
Peshawar	PAK	988005
Baotou	CHN	980000
Adelaide	AUS	978100
Madurai	IND	977856
Mekka	SAU	965700
Köln	DEU	962507
Managua	NIC	959000
Detroit	USA	951270
Shenzhen	CHN	950500

8. **Display Cities Sorted by Name in Ascending Order:** *Scenario:* A geography teacher is preparing a lesson on alphabetical order using city names. You're tasked with providing a sorted list of cities from the database in ascending order by name to support the lesson plan.

```sql
SELECT Name, CountryCode, Population
FROM city
ORDER BY Name ASC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Co

Name	CountryCode	Population
[San Cristóbal de] la Laguna	ESP	127945
´s-Hertogenbosch	NLD	129170
A Coruña (La Coruña)	ESP	243402
Aachen	DEU	243825
Aalborg	DNK	161161
Aba	NGA	298900
Abadan	IRN	206073

9. **Most Populated City:** *Scenario:* A real estate investment firm is interested in cities with significant population densities for potential development projects. You're tasked with identifying the most populated city from the database to guide their investment decisions and strategic planning.

```sql
SELECT Name, CountryCode, Population
FROM city
ORDER BY Population DESC
LIMIT 2;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Cor

Name	CountryCode	Population
Mumbai (Bombay)	IND	10500000
Seoul	KOR	9981619

I have 2 incase the first one is not satisfactory for the shareholders

10. **City Name Frequency Analysis: Supporting Geography Education** *Scenario*: In a geography class, students are learning about the distribution of city names around the world. The teacher, in preparation for a lesson on city name frequencies, wants to provide students with a list of unique city names sorted alphabetically, along with their respective counts of occurrences in the database. You're tasked with this sorted list to support the geography teacher.

```sql
SELECT
    Name,
    COUNT(*) AS NameFrequency
FROM city
GROUP BY Name
ORDER BY Name ASC;
```

Result Grid | Filter Rows: | Expor

Name	NameFrequency
[San Cristóbal de] la Laguna	1
´s-Hertogenbosch	1
A Coruña (La Coruña)	1
Aachen	1
Aalborg	1
Aba	1

11. **City with the Lowest Population:** *Scenario:* A census bureau is conducting an analysis of urban population distribution. You're tasked with identifying the city with

the lowest population from the database to provide a comprehensive overview of demographic trends.

```sql
SELECT Name, CountryCode, Population
FROM city
ORDER BY Population ASC
LIMIT 1;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Co

Name	CountryCode	Population
Adamstown	PCN	42

12. **Country with Largest Population:** *Scenario:* A global economic research institute requires data on countries with the largest populations for a comprehensive analysis. You're tasked with identifying the country with the highest population from the database to provide valuable insights into demographic trends.

```sql
SELECT Name, Population
FROM country
ORDER BY Population DESC
LIMIT 1;
```

Result Grid | Filter Rows: | E

Name	Population
China	1277558000

13. **Capital of Spain:** *Scenario:* A travel agency is organising tours across Europe and needs accurate information on capital cities. You're tasked with identifying the capital of Spain from the database to ensure itinerary accuracy and provide travellers with essential destination information.

```sql
SELECT
    city.Name AS CapitalCity,
    country.Name AS CountryName
FROM country
INNER JOIN city
    ON country.Capital = city.ID
WHERE country.Name = 'Spain';
```

CapitalCity	CountryName
Madrid	Spain

14. **Cities in Europe:** *Scenario:* A European cultural exchange program is seeking to connect students with cities across the continent. You're tasked with compiling a list of cities located in Europe from the database to facilitate program planning and student engagement.

```sql
SELECT
    city.Name AS CityName,
    country.Name AS CountryName,
    city.Population
FROM city
INNER JOIN country
    ON city.CountryCode = country.Code
WHERE country.Continent = 'Europe'
ORDER BY country.Name ASC, city.Name ASC;
```

Name	NameFrequency
[San Cristóbal de] la Laguna	1
´s-Hertogenbosch	1
A Coruña (La Coruña)	1
Aachen	1
Aalborg	1
Aba	1
Abadan	1

15. **Average Population by Country:** *Scenario:* A demographic research team is conducting a comparative analysis of population distributions across countries. You're tasked with calculating the average population for each country from the database to provide valuable insights into global population trends.

```sql
SELECT
    country.Name AS CountryName,
    AVG(city.Population) AS AverageCityPopulation
FROM country
INNER JOIN city
    ON city.CountryCode = country.Code
GROUP BY country.Name
ORDER BY AverageCityPopulation DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

CountryName	AverageCityPopulation
Singapore	4017733.0000
Hong Kong	1650316.5000
Uruguay	1236000.0000
Guinea	1090610.0000
Uganda	890800.0000

16. **Capital Cities Population Comparison:** *Scenario:* A statistical analysis firm is examining population distributions between capital cities worldwide. You're tasked with comparing the populations of capital cities from different countries to identify trends and patterns in urban demographics.

```sql
SELECT city.Name AS CapitalCity,
       country.Name AS CountryName,
       city.Population
FROM City city
INNER JOIN Country country
ON city.ID = country.Capital
ORDER BY city.Population DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Co

CapitalCity	CountryName	Population
Seoul	South Korea	9981619
Jakarta	Indonesia	9604900
Ciudad de México	Mexico	8591309
Moscow	Russian Federation	8389200
Tokyo	Japan	7980230
Peking	China	7472000
London	United Kingdom	7285000
Cairo	Egypt	6789479
Teheran	Iran	6758845

17. **Countries with Low Population Density:** *Scenario:* An agricultural research institute is studying countries with low population densities for potential agricultural development projects. You're tasked with identifying countries with sparse populations from the database to support the institute's research efforts.

```sql
SELECT Name,
       Population,
       SurfaceArea,
       (Population / SurfaceArea) AS PopulationDensity
FROM Country
WHERE (Population / SurfaceArea) < 50
ORDER BY PopulationDensity ASC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Name	Population	SurfaceArea	PopulationDensity
French Southern territories	0	7780.00	0.0000
Antarctica	0	13120000.00	0.0000
Heard Island and McDonald Islands	0	359.00	0.0000
Bouvet Island	0	59.00	0.0000
British Indian Ocean Territory	0	78.00	0.0000
South Georgia and the South Sandwich Islands	0	3903.00	0.0000
United States Minor Outlying Islands	0	16.00	0.0000
Greenland	56000	2166090.00	0.0259

18. **Display Columns with Limit (Rows 31-40):** *Scenario:* A market research firm requires detailed information on cities beyond the top rankings for a comprehensive analysis. You're tasked with providing data on cities ranked between 31st and 40th by population to ensure a thorough understanding of urban demographics.

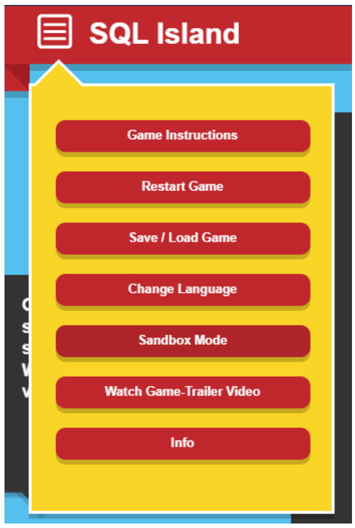
```sql
SELECT Name, CountryCode, Population
FROM City
ORDER BY Population DESC
LIMIT 10 OFFSET 30;
```

Result Grid | Filter Rows: | Export:

Name	CountryCode	Population
Shenyang	CHN	4265200
Kanton [Guangzhou]	CHN	4256300
Singapore	SGP	4017733
Ho Chi Minh City	VNM	3980000
Chennai (Madras)	IND	3841396
Pusan	KOR	3804522
Los Angeles	USA	3694820
Dhaka	BGD	3612850

Additional Resources

1) SQLBolt - Learn SQL - Introduction to SQL : Welcome to SQLBolt, a series of interactive lessons and exercises designed to help you quickly learn SQL right in your browser.
2) MySQL Tutorial: Tutorial website.
3) MySQL Basics Tutorial Series - YouTube: MySQL Basics Tutorial Series
4) Hackerrank Challenges: How to use hackerrank challanges.odt
5) Sakila DB Exercises:
   a. Use this file to create the schema: create sakila db.sql
   b. Access exercises here: sakila db exercises combined.sql
   c. Follow each step to create your database and upload the exercise file. **how to create sakila db and upload exercises.webm**



6) SQL Island: After the survived plane crash, you will be stuck on SQL Island for the time being. By making progress in the game, you will find a way to escape from this island.
   Change Language to English.
7) You can watch the video on how to open SQL file:How to create a schema in workbench.webm

Course Notes

It is recommended to take notes from the course, use the space below to do so, or use the revision guide shared with the class:

We have included a range of additional links to further resources and information that you may find useful, these can be found within your revision guide.

END OF WORKBOOK

Please check through your work thoroughly before submitting and update the table of contents if required.

Please hand in your workbook in MS Assignment page.